

# Preliminary Research Note

Problem framing, evidence map, assumptions, and early design direction

PROJECT

## Stator and Rotor Dynamics Analysis

|             |                      |          |                              |
|-------------|----------------------|----------|------------------------------|
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| Author      | NervTech Research    | Class    | INTERNAL                     |

### Research Question

Which actuator architecture gives the best path to a compact, safe, and controllable robotic joint prototype?

### Abstract

This document captures early research for the Integrated Robotic Actuator project. It is intended to separate observations, assumptions, risks, and design direction before committing to a detailed technical design.

**Keywords:** integrated actuator, robotics, torque density, control architecture, preliminary research

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## 1 RESEARCH SNAPSHOT

### Research Snapshot

|                         |                                                                                                                                                                    |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Problem</b>          | Build a compact integrated robotic actuator that combines motor, reduction, sensing, drive electronics, and control firmware.                                      |
| <b>Key Finding</b>      | The highest leverage early decision is the actuator architecture, because it constrains torque density, controllability, thermal behaviour, and manufacturability. |
| <b>Main Uncertainty</b> | The true operating envelope is still uncertain: load profile, duty cycle, heat rejection, encoder resolution, and gearbox backlash all need validation.            |
| <b>Next Step</b>        | Prototype the smallest credible bench rig and measure torque, speed, temperature rise, backlash, and closed-loop response.                                         |

## 2 PROBLEM FRAMING

### Research Question – System Direction

What architecture can demonstrate a credible portfolio-grade actuator while still being feasible with undergraduate-level tools, budget, and testing equipment?

The preliminary research phase should not over-specify the product. Its job is to reduce uncertainty and identify the design direction that is most likely to produce a demonstrable robotic actuator prototype.

### Hypothesis – Architecture

A modular integrated actuator with a BLDC motor, compact reduction stage, magnetic encoder, current sensing, and embedded field-oriented control is likely to provide the best balance between technical ambition and achievable implementation.

## 3 EVIDENCE MAP

Table 1: Evidence Matrix

| Claim                            | Evidence / Observation                                                                                                      | Source                 | Confidence |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------|------------------------|------------|
| Integrated packaging is valuable | Combining actuation, sensing, and drive control reduces wiring complexity and makes the unit easier to reuse across robots. | Project observation    | Medium     |
| Backlash matters                 | Low backlash is important for repeatable joint control and credible closed-loop demonstrations.                             | Bench-test requirement | High       |
| Thermal limits may dominate      | Small actuators can reach thermal limits before electrical or mechanical limits during sustained load.                      | Engineering assumption | Medium     |

## 4 ASSUMPTIONS TO VALIDATE

Table 2: Assumptions to Validate

| Assumption                     | Why it matters                                                      | Validation method                                       | Owner      |
|--------------------------------|---------------------------------------------------------------------|---------------------------------------------------------|------------|
| Motor torque is sufficient     | Sets whether the reduction ratio and motor frame size are credible. | Measure no-load speed and loaded torque on a bench rig. | Controls   |
| Encoder resolution is adequate | Limits position accuracy and velocity estimation quality.           | Compare commanded and measured step response.           | Firmware   |
| Thermal rise is manageable     | Determines continuous operating limits.                             | Run fixed-duty thermal test with logging.               | Electrical |

## 5 OPEN GAPS

### Open Gap – Unknowns

The research note should explicitly track what is not known yet: final motor choice, gearbox type, torque target, power supply envelope, controller current limit, and safety cut-off design.

### Risk – Prototype Risk

The main early risk is selecting parts before the test envelope is clear. Keep the first rig instrumented and conservative, especially around power electronics and rotating hardware.

## 6 RESEARCH BACKLOG

Table 3: Research Backlog

| ID    | Question / Task                                                                                               | Output                   | Priority |
|-------|---------------------------------------------------------------------------------------------------------------|--------------------------|----------|
| R-001 | Compare candidate actuator architectures: direct drive, belt reduction, planetary gearbox, cycloidal concept. | Architecture trade study | High     |
| R-002 | Build a first-order torque-speed and thermal model.                                                           | Notebook with plots      | High     |
| R-003 | Define safety plan for power bring-up and moving-mechanism tests.                                             | Bring-up checklist       | High     |
| R-004 | Identify demo scenarios for the final portfolio video.                                                        | Demo storyboard          | Medium   |

## 7 PRELIMINARY DIRECTION

### Direction / Decision – Next Iteration

Proceed toward a bench-testable integrated actuator module. The next document should convert this research into measurable requirements, architecture diagrams, and a verification plan.